Probability and Statistics for Computing Science

STAT-271

Spring 2025

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1. Descriptive Statistics

- Statistics is the art of learning from data.
- Statistics is used to *evaluate* observations and *make decisions*.
- Examples of observations:
  - Average temperature over many years
  - Unemployment rate
  - Rate of inflation
  - Goals scored by a player in each game
  - Work productivity
  - Happiness index
  - Test scores of students

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Typically, a statistical analysis begins with a statement (or a question) that will be addressed by studying some data.

Two “types” of statistics, or procedures:

- **Descriptive statistics**
  - Describe, organize, summarize, analyze data
  - Numerical summaries
  - Graphs
- **Inferential statistics**
  - Interpret the meaning of information
  - Generalize the results or characteristics of a sample (or experiment) to a larger group
  - Draw conclusions, answer a question, make a decision

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1.1 Data Collection

- Observational data
  - Information collected without the researcher's intervention
  - Surveys are one kind of observational studies
  - Observe and measure variables of interest
  - No attempt to influence the responses/observations
  - The main goal is to describe the characteristics of the dataset
  - See the examples on slide 2
- Experimental data
  - Information collected from an appropriate experiment
  - Impose a "treatment" and study the response/effect on a variable
  - The design of the experiment depends on the use that one wants to make of the data

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Example of an experiment:

- Study the effectiveness of two *teaching methods* (**independent variable**).
- Divide the students into two groups.
- Use a different teaching method for each group.
- Compare the test scores of each group to determine if the *scores* are significantly higher in one group (**dependent variable**).
- If so, it might be reasonable to assume that one teaching method is superior.

An **essential condition** to draw a valid conclusion:

- The two groups were selected "*at random*".
- All possible choices of the members of a group are equally likely.

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1.2 Statistical inference

- To draw logical conclusions from data
- Must take into account the possibility of chance
- Usually make some assumptions about the probabilities of obtaining different data values
- The totality of these assumptions is referred to as the probability model
- The nature of the data often suggests appropriate probability models

Example:

- Proportion of new computer chips that are defective
- Select some chips randomly and test them
- A reasonable model is that each chip is defective with probability p (Bernoulli random variable)

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1.3 Population versus Sample

Statistics is about obtaining information about all "*individuals*".

The set of all "individuals" is referred to as the **population.**

It is common for the population to be too large to study all individuals. In such cases, a subgroup of the population, called a **sample**, is studied and results are used to understand the population.

To be *informative*, a sample should be *representative* of the population.

A sample is generally considered not representative unless it was chosen "*at random*".

Choosing a sample according to a non-random rule generally introduces *bias* in the observations, i.e. a systematic distortion in favor of certain values.

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Example of a sample:

Suppose you are interested in the age distribution of people residing in Vancouver.

You record the ages of the first 50 people to go to a popular gym and find that the average age of these 50 people is 30.4.

Can you conclude that the average age of people residing in Vancouver is approximately 30.4? Explain.

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1.4 Datasets

The data collected from a study should be presented in a clear and concise way, revealing the important characteristics of the data.

We are generally interested in features such as the location, the range, the degree of concentration, the symmetry of the data.

Tables and/or *graphs* can be used to present data.

Which tables and graphs are commonly used depends on

- Type of observations: numerical values or not
- Size of dataset: relatively few distinct values or large number of different values that may be grouped into intervals

Numerical data can also be summarized using common "*statistics*".

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1.4.1 Describing data

A ***frequency table*** can be used to present a dataset with relatively few distinct values.

For example, the starting yearly salaries of 42 recently graduated students are displayed in a frequency table.

Starting Salary ('000s)	Frequency
57	4
58	1
59	3
60	5
61	8
62	10
63	0
64	5
65	0
66	2
67	3
70	1

Source: Table 2.1

Introduction to Probability and Statistics for Engineers and Scientists, 6th Ed, Ross

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The dataset of salaries can be displayed graphically in a ***line graph***.

The distinct values appear on the x-axis and the heights of the bar indicates the frequencies.

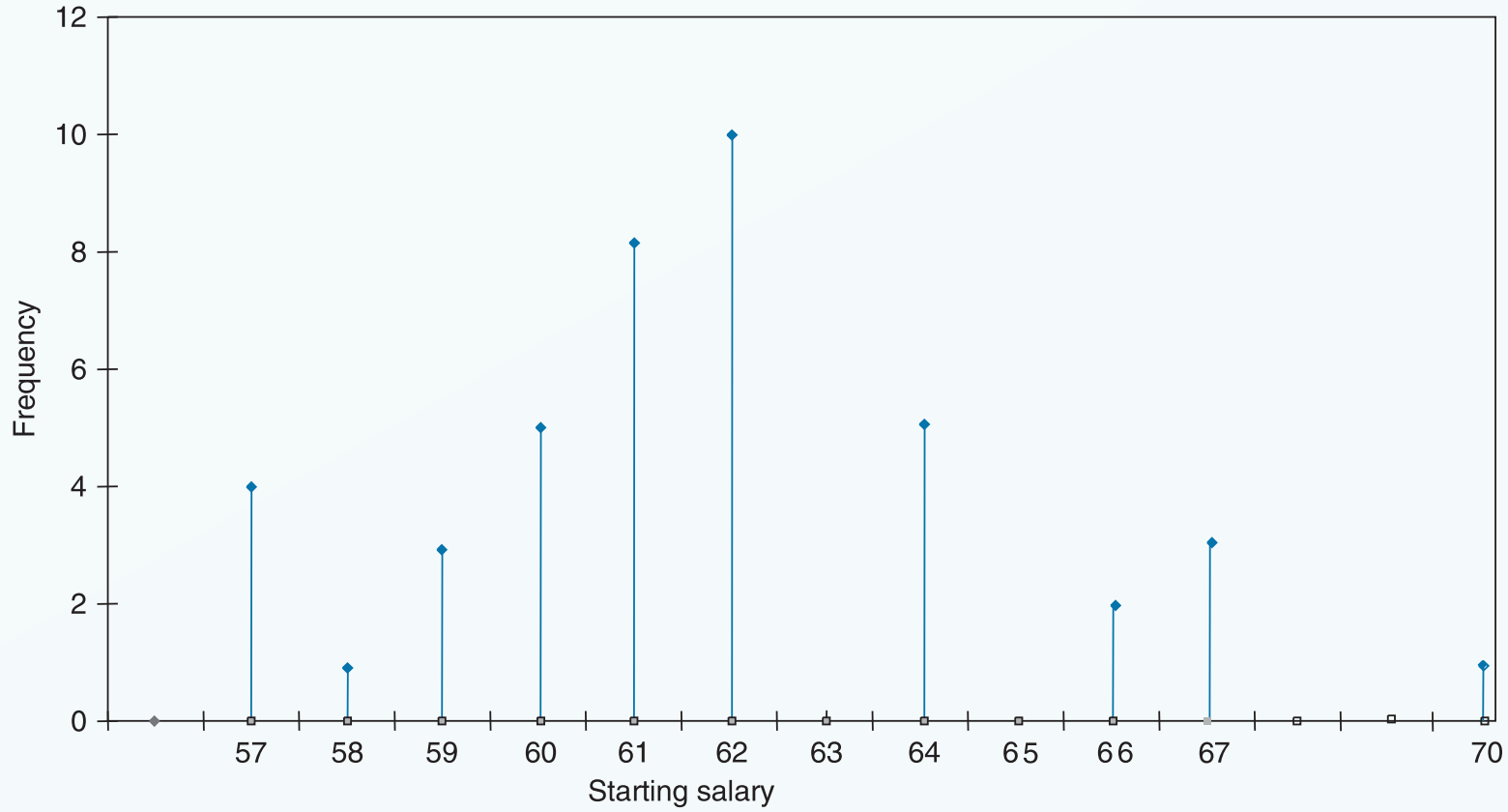


FIGURE 2.1 Starting salary data.

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Instead of using lines, it is common to use rectangles in a ***bar graph***.

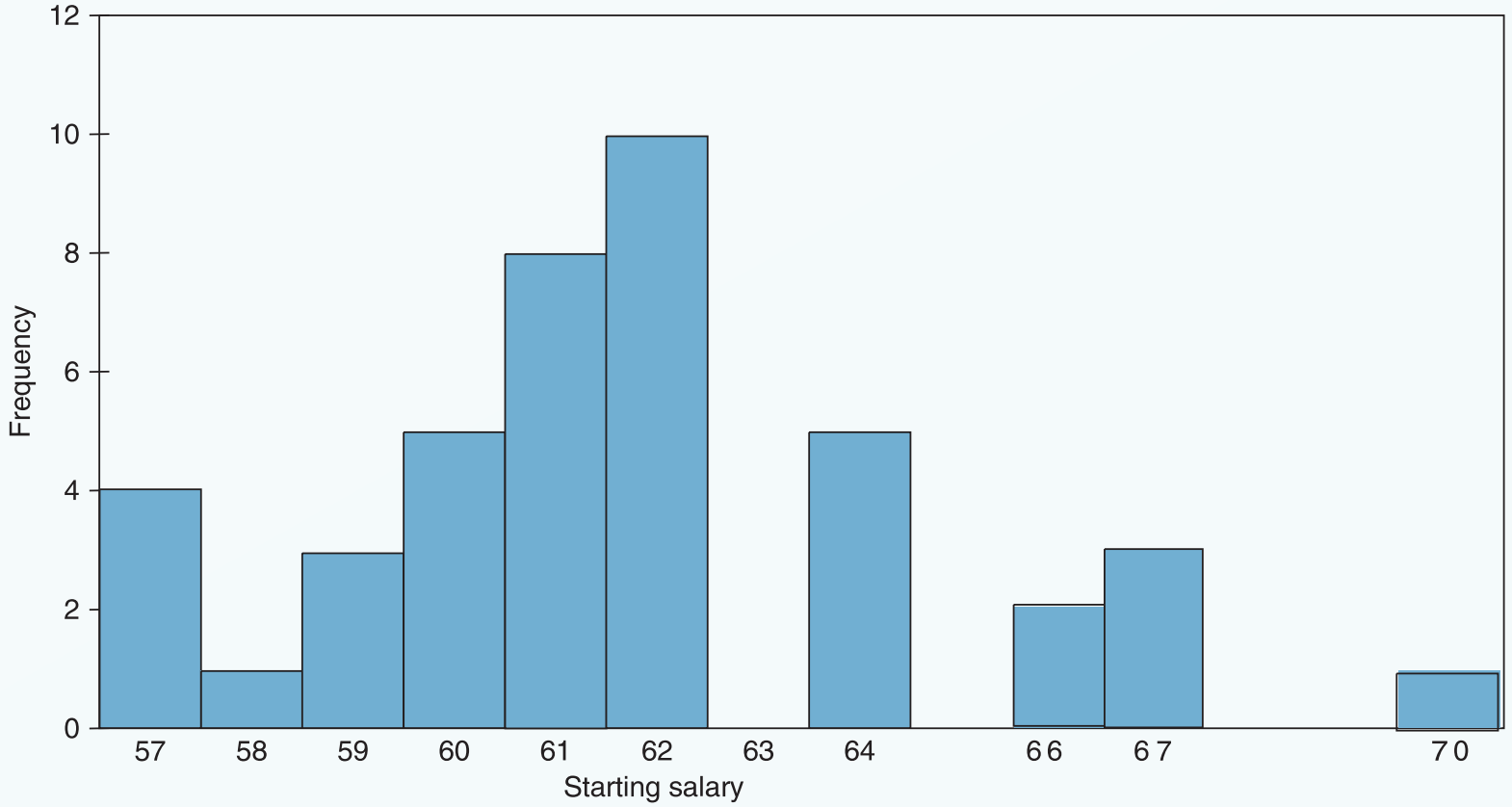


FIGURE 2.2 Bar graph for starting salary data.

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The ***frequency polygon*** is another type of graph which is obtained by joining the frequencies in a line graph.

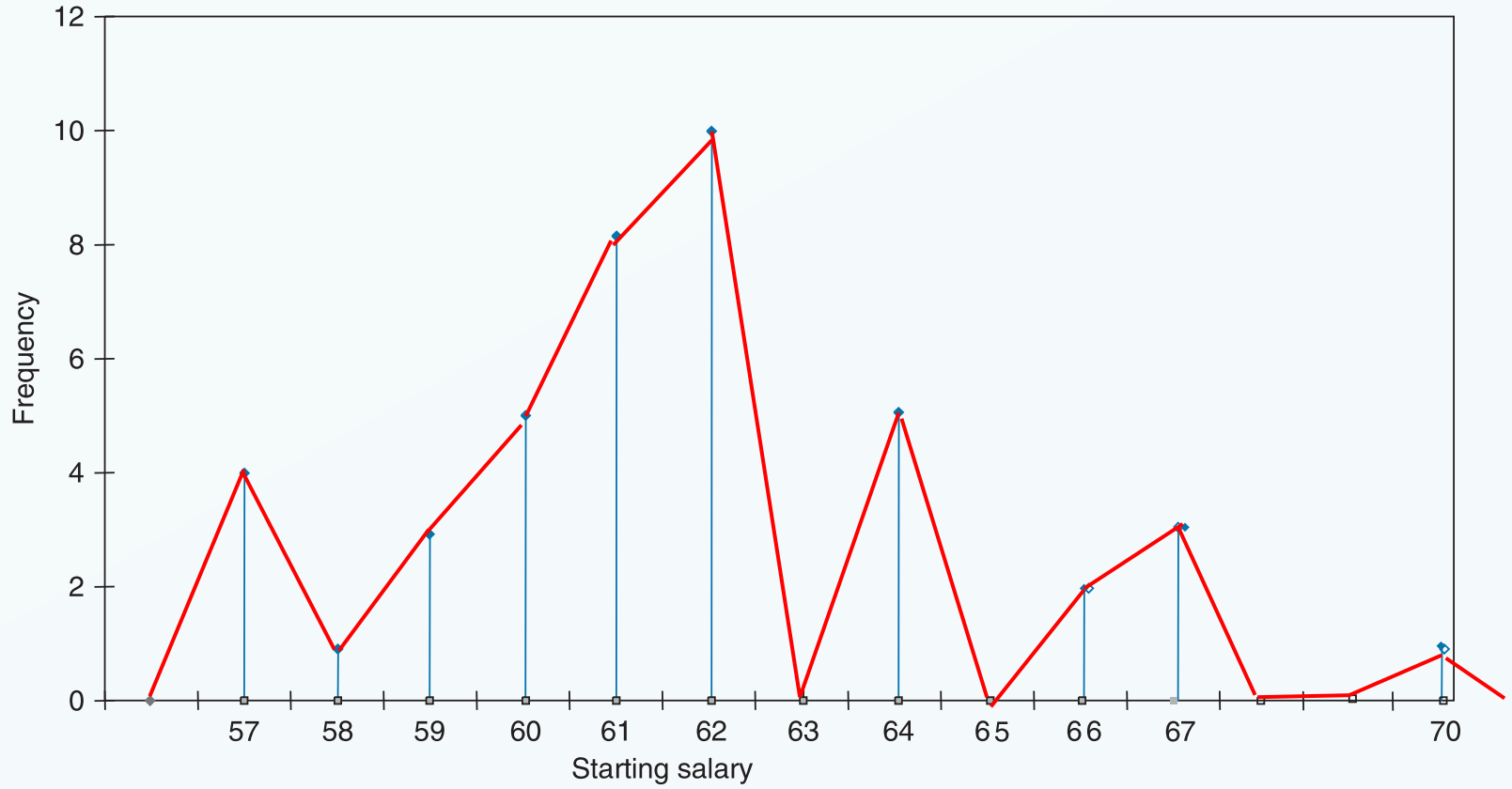


FIGURE 2.3 Frequency polygon for starting salary data.

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The *relative frequency table* for the starting salary dataset is

Starting Salary ('000s)	Relative Frequency
57	4/42=0.0952
58	1/42=0.0238
59	3/42
60	5/42
61	8/42
62	10/42
63	0/42
64	5/42
65	0/42
66	2/42
67	3/42
70	1/42

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When the data is not numerical, a ***pie chart*** is often used to show the relative frequencies for each distinct value. The area of a sector is proportional to the relative frequency of a given value.

The following graph indicates

the relative frequencies of different

cancers affecting 200 patients.

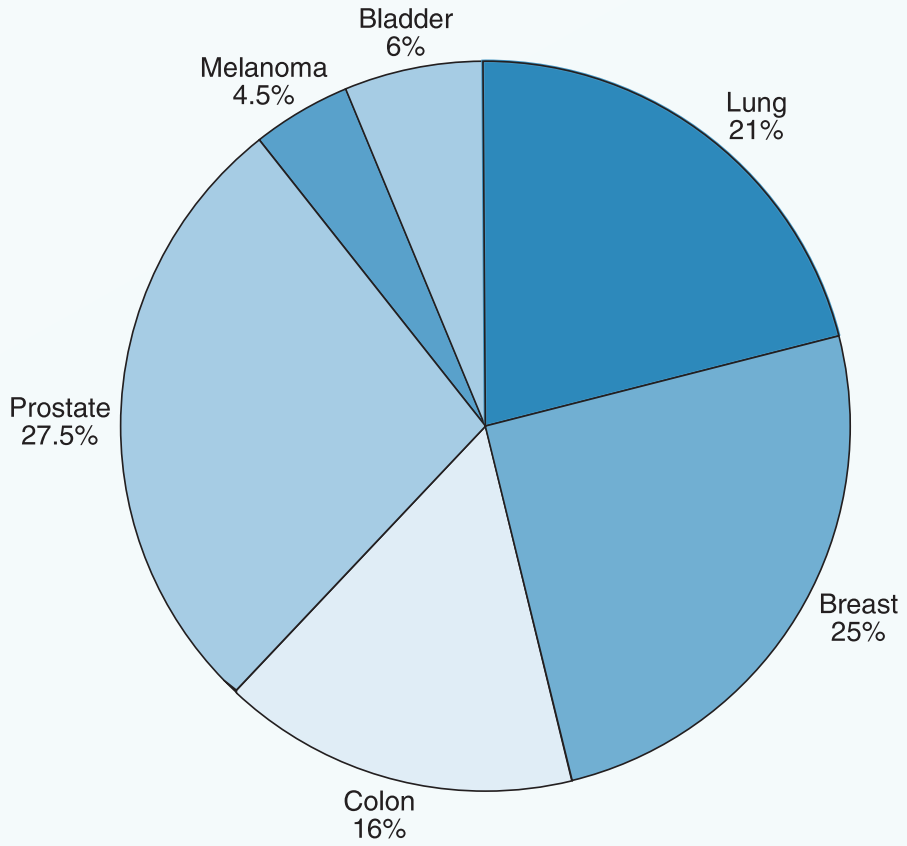


FIGURE 2.4

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1.4.2 Describing data – Grouped data

When the dataset contains a large number of distinct values, a more effective way to display the data is to group them into intervals.

Considerations when choosing the class intervals:

-Few classes: risk losing too much relevant information

-Many classes: results in low frequencies and difficulty identifying patterns

-Commonly, 5 to 10 classes are appropriate (but the number of classes is a subjective choice)

-Class intervals of equal length are common but unequal lengths may be used

-*Class boundaries*: the endpoints of an interval

Usually the lower endpoint is included in the interval but not the upper endpoint (left-end inclusion convention)

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Table 2.3 of the textbook lists the lifetimes of 200 incandescent lamps.

Using class intervals of length 100, starting at 500 hours, the *class frequency table* is

Table 2.4

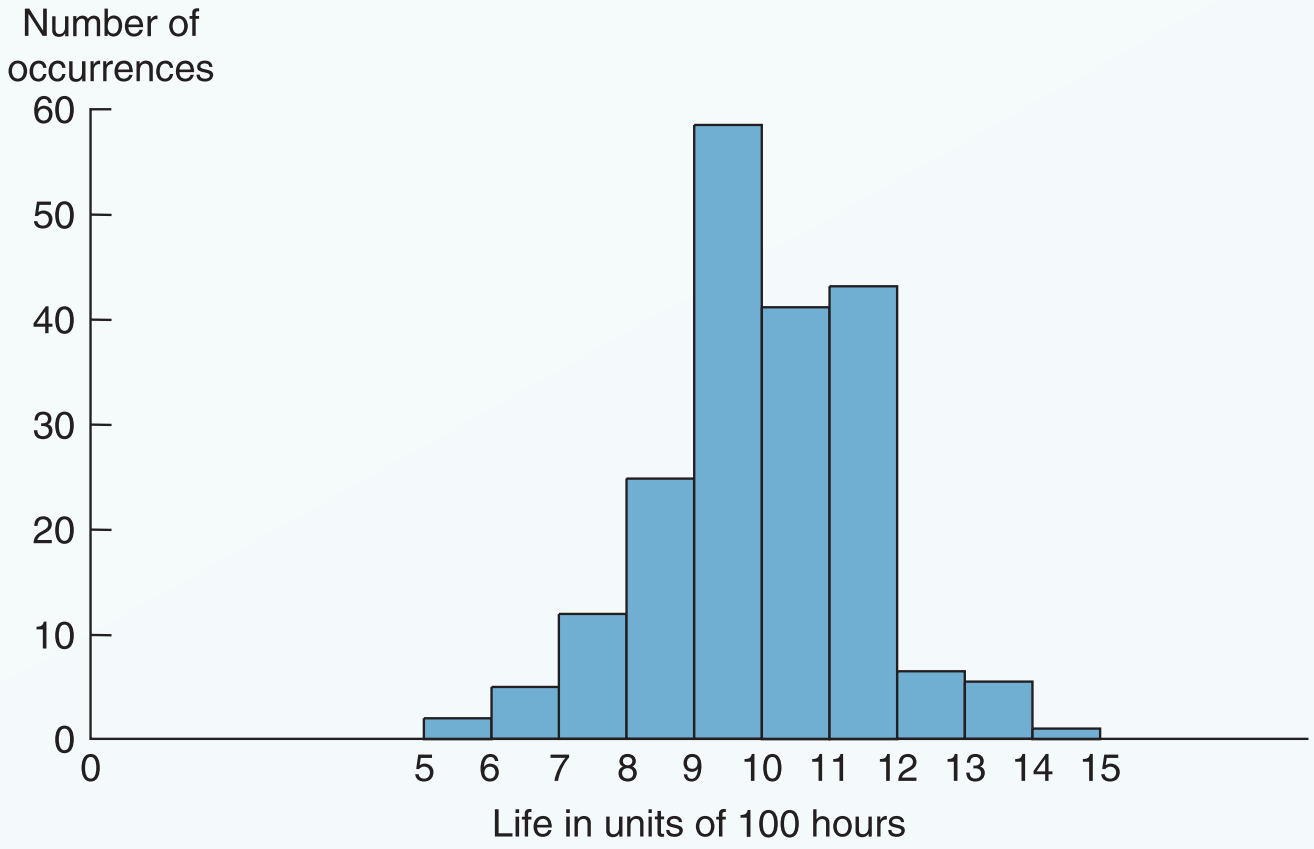
Class Interval (hours)	Frequency
500-600	2
600-700	5
700-800	12
800-900	25
900-1000	58
1000-1100	41
1100-1200	43
1200-1300	7
1300-1400	6
1400-1500	1

So, the lifetime of 2 lamps was greater or equal to 500 and less than 600

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A ***histogram*** is a bar graph with the bars adjacent to each other.

The frequency histogram for the data in Table 2.4 is



A relative frequency histogram would show the relative frequencies on the y-axis.

FIGURE 2.5 A frequency histogram.

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An ***ogive*** is a cumulative (relative) frequency plot. It plots the proportion of values less than or equal to each value on the x-axis.

The ogive for the data in Table 2.4 is

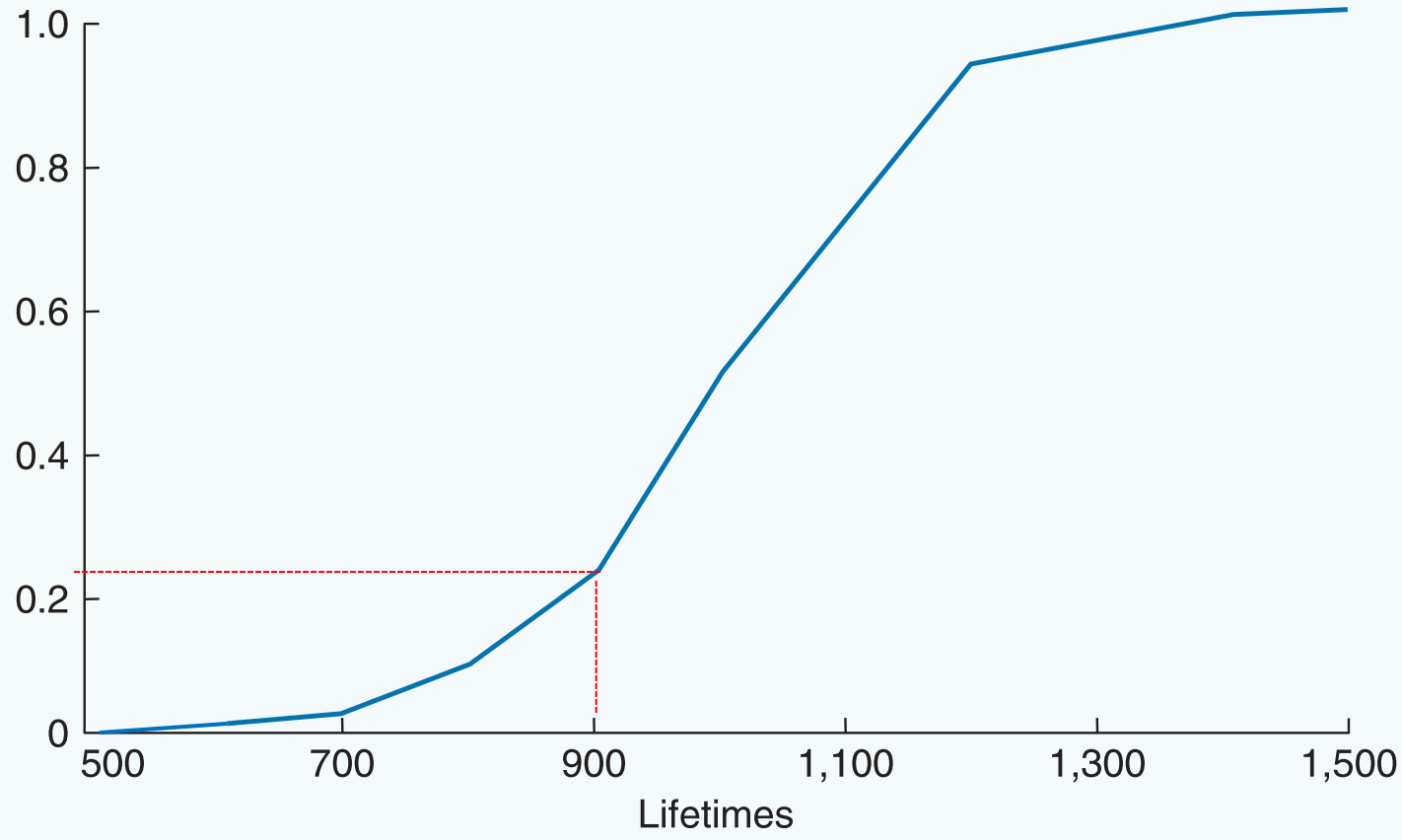


FIGURE 2.6 A cumulative frequency plot.

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A ***stem and leaf plot*** can be an efficient way to display a small to moderate-sized dataset.

Each value is divided into two parts, a stem and a leaf.

Example: Display the following values in a stem and leaf plot.

24, 25, 31, 33, 34, 34, 36, 38, 39, 42, 44, 47, 51, 58, 63

2 4,5

3 1,3,4,4,6,8,9

4 2,4,7

5 1,8

6 3

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1.4.3 Summarizing data

The information in large datasets can be summarized by suitable measures called statistics. A ***statistic*** is a numerical quantity whose value is determined by the data.

Central tendency

Some statistics are used to summarize the "location" of a set of values.

In order to define such statistics, we will suppose that we have n numerical values $x_1, x_2, \ldots, x_n$.

Sample Mean (average)

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

The sample mean is the arithmetic average of all values.

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When using a frequency table with k distinct values v_i for i=1,2,…,k and corresponding frequencies f_i, the sample mean of the n values is

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{k} f_i \, v_i$$

f_i: frequency of value v_i for i=1,2,…,k

$\sum_{i=1}^{k} f_i = n$

It is a weighted average of the k distinct values.

Example: Calculate the sample mean of 20 values if the frequency table is

Values, v	Frequency, f
10	2
15	5
20	8
25	5

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Exercise:

Let $y_i = ax_i + b$. Show that $\overline{y} = a\overline{x} + b$.

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Sample Median

$$\tilde{x} = \begin{cases} x_{\left(\frac{n+1}{2}\right)} & n\ odd \\ 0.5\left(x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}\right) & n\ even \end{cases}$$

where n is the number of values in the dataset

$x_{(i)}$ is the ith smallest value when ordered from smallest to largest

The sample median is the "middle value" of the dataset.

Example: Calculate the sample median of 20 values if the frequency table is

Values, v	Frequency, f
10	2
15	5
20	8
25	5

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Remarks:

-The sample mean uses all values and is affected by extreme values (large or small).

-The sample median uses only “middle” values and is not affected by extreme values.

The ***sample mode*** is another measure of central tendency.

It is the value that occurs with the largest frequency.

If more than one value occurs most frequently, those values are called *modal values*.

Example: Calculate the sample mode of 20 values if the frequency table is

Values, v	Frequency, f
10	2
15	5
20	8
25	5

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Dispersion

A statistic used to summarize the "dispersion" of a set of values is the **standard deviation**. It measures the spread or variability of a set of values.

The *sample variance*, denoted s^2, of n numerical values $x_1, x_2, \ldots, x_n$, is

$$\boldsymbol{s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

It is the average of the squares of the distances between the values and the mean.

The sample standard deviation is s, the positive square root of the variance.

The unit of the standard deviation is the same unit as the data. The unit of the variance is the unit of the data squared.

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Example: Calculate the sample standard deviations of the two datasets:

Dataset 1: 8, 8, 9, 10, 11, 14m

Dataset 2: 6, 8, 10, 11, 15m

Dataset 1: $\bar{x} =$

$\sum(x_i - \bar{x})^2 =$

$s^2 =$ and $s =$

Dataset 2: $\bar{x} =$

$\sum(x_i - \bar{x})^2 =$

$s^2 =$ and $s =$

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Exercise: Show that $\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n}(x_i)^2 - n\bar{x}^2$

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Percentiles

The **sample 100*p* percentile** is the value that satisfies

i- at least 100p percent of the values are less or equal to it;

ii- at least 100(1-*p*) percent of the values are greater or equal to it.

If two values satisfy these conditions, the 100*p* percentile is the average of these two values.

$$100p \text{ percentile} = \begin{cases} x_{(\lfloor np+1 \rfloor)} & np \text{ is not an integer} \\ 0.5\left(x_{(np)} + x_{(np+1)}\right) & np \text{ is an integer} \end{cases}$$

where $x_{(\mathrm{i})}$ is the ith smallest value and $\lfloor i \rfloor$ is the integer part of i .

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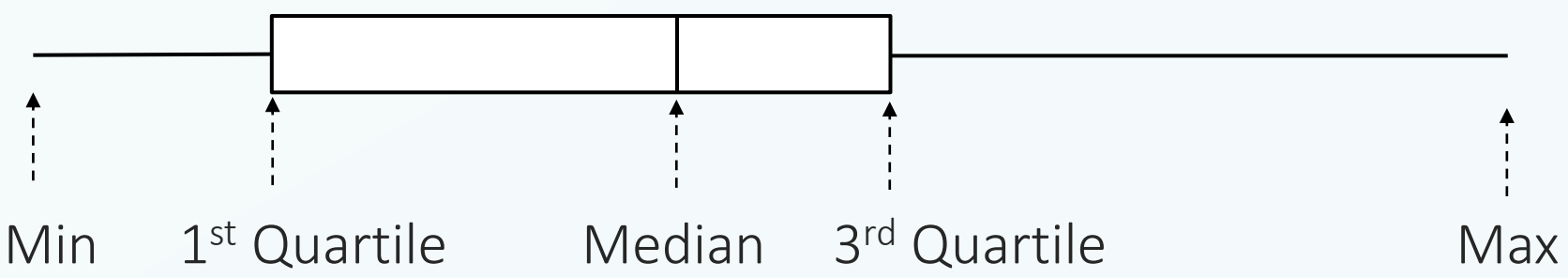
Special cases:

-First quartile: 25 percentile

-Second quartile: median or 50 percentile

-Third quartile: 75 percentile

A box plot summarizes key statistics of a dataset.



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Example: Find the quartiles and plot the box plot of the following dataset.

25, 26, 29, 32, 34, 34, 35, 37, 38, 40, 42, 46, 55, 68

1st quartile:

Median:

3rd quartile:

Box plot:

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1.5 Normal Datasets and Skewed Datasets

Some large datasets have histograms that are "bell-shaped". They are called **normal** datasets.

Their histograms peak at the sample median (also equal to the sample mode and sample mean) and they are symmetric.

When the histogram is *not symmetric* about its median, it is said to be *skewed*.

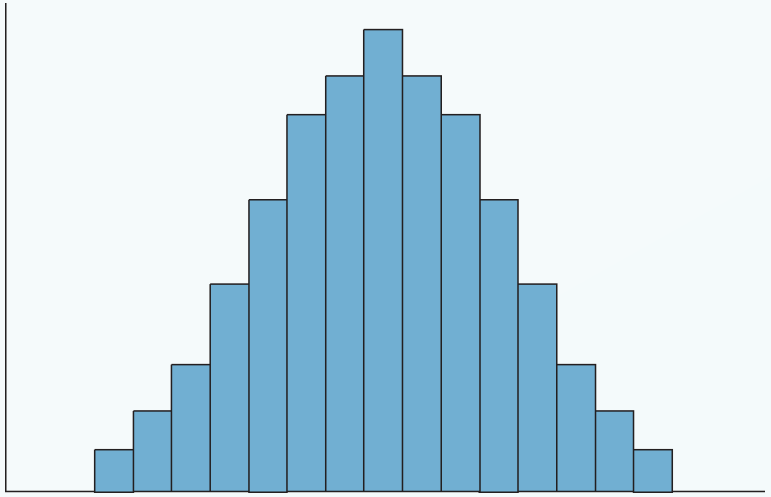


FIGURE 2.8 Histogram of a normal data set.

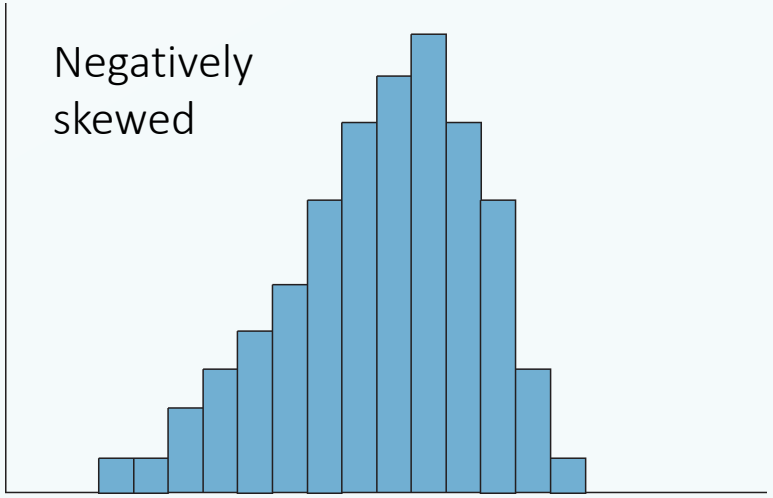


FIGURE 2.10 Histogram of a data set skewed to the left.

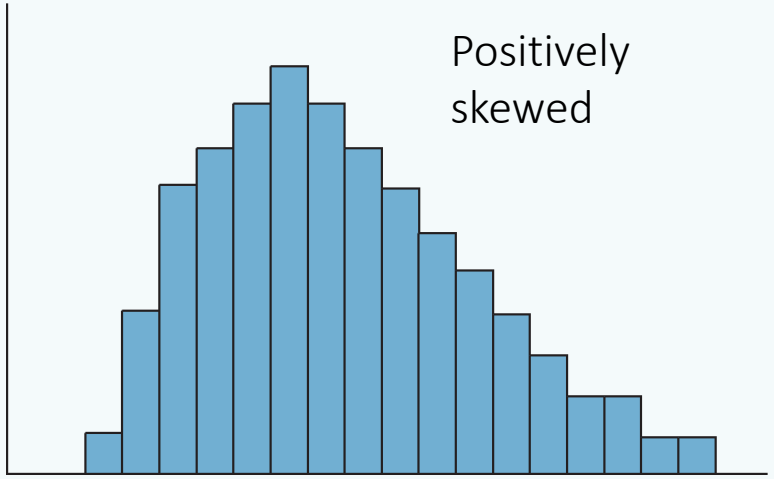


FIGURE 2.11 Histogram of a data set skewed to the right.

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1.6 Paired datasets and correlation

Some datasets consist of pairs of values that are related. The i[th] data value can be represented by (x_i, y_i). Such dataset can be displayed in a **scatterplot**, a 2-dimensional graph with the y values plotted against the x values.

A statistic that measures the linear relationship between the paired values is the **sample correlation coefficient, *r***.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y}$$

Some properties of r:

i. $-1 \leq r \leq 1$

ii. r is unitless

iii. Linear transformations of the values: $r_{a+bx,c+dy} = sign(bd)\, r_{x,y}$

iv. If $y_i = a + bx_i$; with b<0, then $r = -1$ and with b>0, $r = 1$

$$sign(a) = \begin{cases} -1 & a < 0 \\ 0 & a = 0 \\ 1 & a > 0 \end{cases}$$

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Examples of scatterplots:

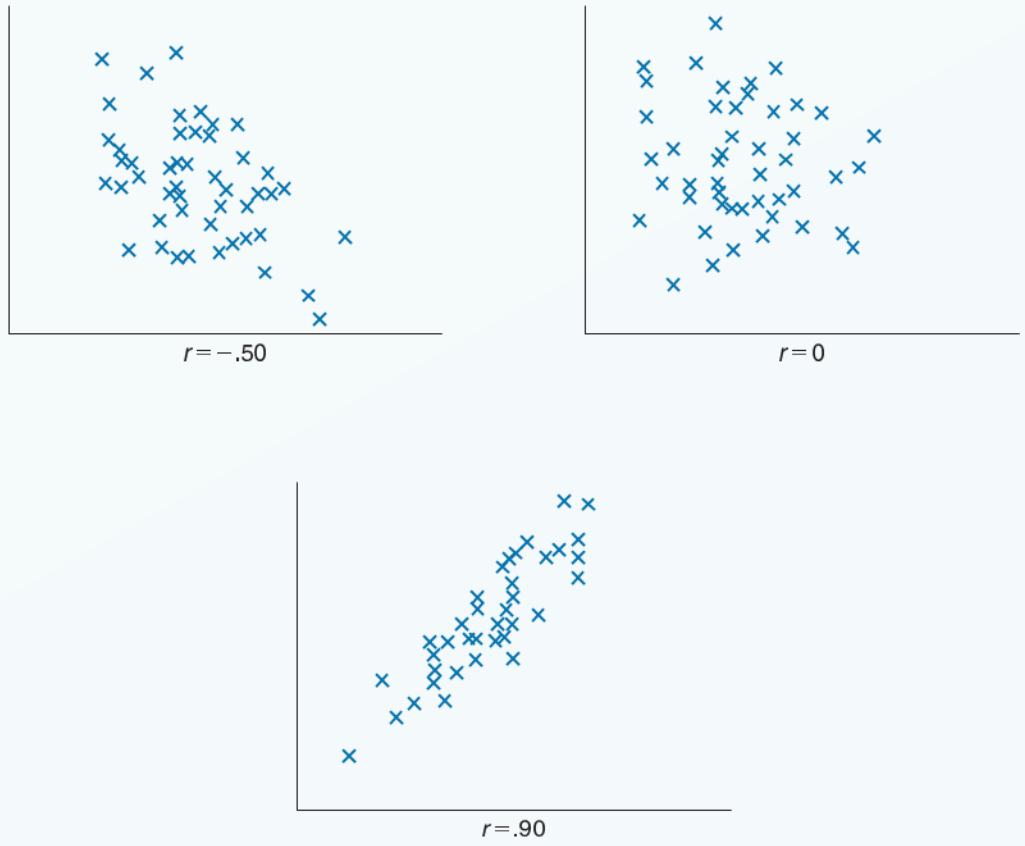


FIGURE 2.14 Sample correlation coefficients.

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Remarks:

i. The correlation coefficient is a measure of the linear relationship between paired values.

Consider the paired values: (-2, 4), (-1, 1), (0, 0), (1, 1) and (2, 4). Clearly, the y values are related to the x values.

However, the correlation coefficient is $r = 0$.

ii. Correlation measures association, not causation.

A study might show that owners of two cars tend to live longer than owners of one car. This would result in a positive correlation between the number of cars and the lifetimes. However, there is no cause-and-effect relationship between the number of cars and the lifetime. (Buying a second car is not likely to increase one's lifetime.)

There is a positive correlation between obesity in parents and their children. Given that body type is hereditary and that children may adopt their parents' habits, it is reasonable to assume that there is a direct causal link here.

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